

CS 5542 — BIG DATA ANALYTICS

Controlled E-Commerce Product Image Generation

Using Stable Diffusion XL for Structured Prompt Engineering &
Reproducible Workflows

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Project Vision

Standardizing AI-driven product catalogs at scale

THE CHALLENGE & THE GOAL

The Problem

- "Hallucinations" in generic AI prompts
- Random and cluttered backgrounds
- Inconsistent lighting and styles
- High SKU variance during generation

The Solution

A pipeline that transforms structured product metadata into catalog-ready, professional studio photography with 100% background consistency.

STRUCTURED PRODUCT DATASET

Category	Products Included
Apparel	Women Summer Dress, Hoodie
Footwear	Running Shoes, Sneakers
Accessories	Leather Backpack, Wrist Watch, Handbag
Electronics	Smart Speaker, Wireless Earbuds

Scale: 10 Products × 3 Seeds × 2 Strategies = 60 Images

Attributes: product_id, title, category, color, material, style

PIPELINE ARCHITECTURE

- ✓ **Seed-Controlled:** Ensures exact reproducibility.
- ✓ **Performance:** fp16 precision on A100 GPU.
- ✓ **PromptEngine:** Maps CSV metadata to prompts.
- ✓ **Automated Eval:** CLIP Score & SSIM calculation.



PROMPT ENGINEERING STRATEGIES

STRATEGY A: NAIVE

Baseline

"Coffee Mug"

Results in unpredictable lighting, messy backgrounds, and non-pro styles.

STRATEGY B: STRUCTURED

Improved

"Professional studio photography of a white ceramic Coffee Mug, minimalist aesthetic, white solid background..."

Controlled background, standardized lighting, high-end feel.

TECHNICAL STACK & TOOLS



SDXL 1.0

1024x1024 Base resolution



Diffusers

HF Pipeline management



CLIP/SSIM

Alignment & Consistency metrics



Google Colab

A100 GPU Integration

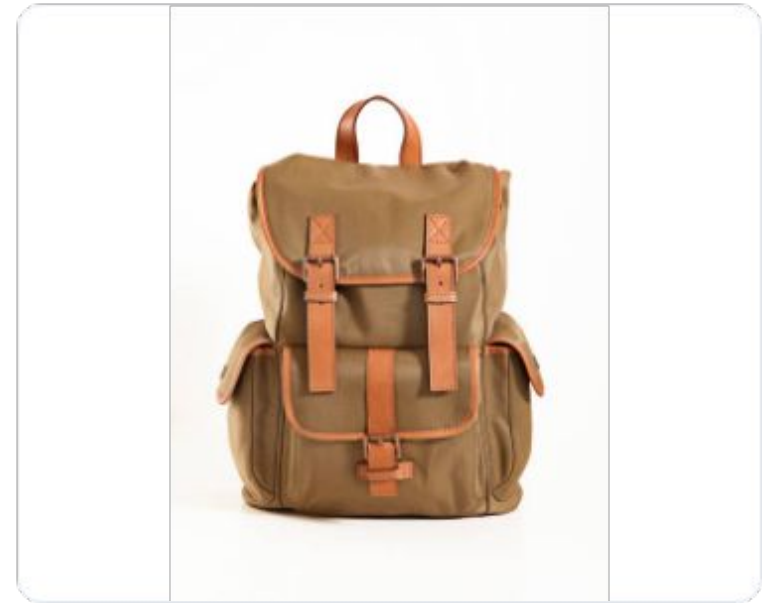
VISUAL RESULTS: STRUCTURED CONSISTENCY



Coffee Mug



Sneakers

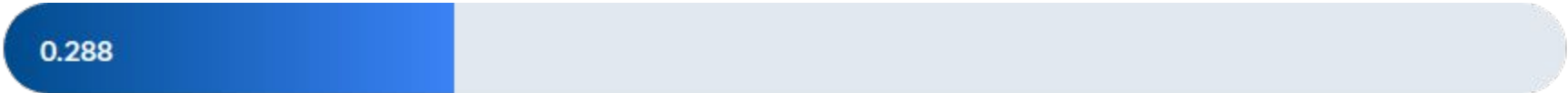


Leather Backpack

CLIP SCORE: SUBJECT ALIGNMENT

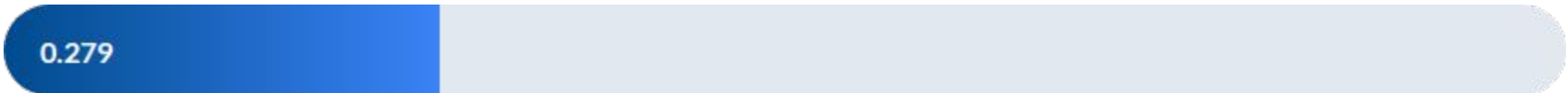
Baseline (Naive)

0.288



Structured (Improved)

0.279

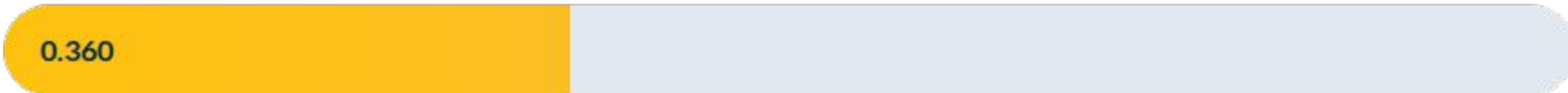


Insight: Alignment is statistically equivalent. Adding control tokens does NOT confuse the model's subject recognition while standardizing the aesthetic.

SSIM: VISUAL CONSISTENCY GAIN

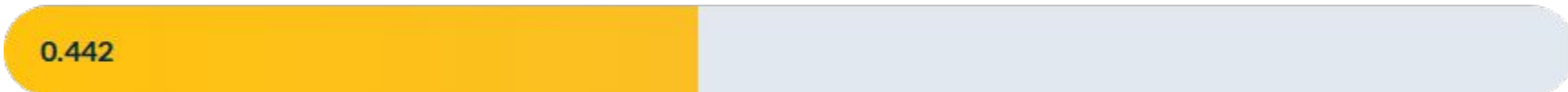
Baseline SSIM

0.360



Improved SSIM

0.442



+22.8%

Improvement in structural similarity across seeds

FAILURE ANALYSIS & LIMITATIONS

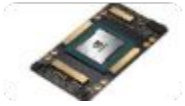
- ✔ **Component Drift:** Products with straps (backpacks) fluctuated placement.
- ✔ **Geometric Complexity:** Footwear laces showed high variance across seeds.
- ✔ **Fine Detail:** Watch dials often blurry; requires ESRGAN Super-Resolution.
- ✔ **Variety Trade-off:** High consistency reduces "creative" background options.

Questions?



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IMAGE SOURCES



<https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/a100/nvidia-a100-hgx-3qtr-front-left-2c50-p.jpg>

Source: www.nvidia.com



https://static.vecteezy.com/system/resources/previews/068/330/619/large_2x/two-floating-white-coffee-mugs-with-minimalist-design-and-soft-lighting-photo.jpg

Source: www.vecteezy.com



<https://static.vecteezy.com/system/resources/thumbnails/049/668/138/small/nike-air-max-exceed-running-shoes-orange-and-blue-comfortable-and-stylish-with-a-transparent-background-png.png>

Source: www.vecteezy.com



<https://media.istockphoto.com/id/1132979378/photo/canvas-backpack-isolated-on-white-background.jpg?s=612x612&w=0&k=20&c=kZlwQfUBZgL9efZm93Dr2dPY3RYcAZ1xJe1XJV0dm4=>

Source: www.istockphoto.com